

Company Knowledge AI Assistant

Final Project · Option 4 · Petalnex AI Automation Internship

A RAG-powered assistant that answers internal company questions using source-aware retrieval, safe fallback, validation, storage, escalation, and memory.



**Ask → Retrieve →
Answer**

Knowledge base · Gemini ·
Supabase · n8n

20+

knowledge
entries

15/15

tests passed

Presentation Roadmap

A fast walkthrough from business need to implementation, testing, and demo.

01

Problem & current process

Why this project?

02

Proposed solution

How the assistant works

03

Architecture & workflows

Core + intelligent layers

04

AI/RAG integration

Grounded answers + fallback

05

Reliability & security

Handling failure safely

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Demo & outcomes

What I will show live

Business Problem

Employees waste time searching scattered company information or waiting for manual answers.

Existing Process

- Policies and FAQs stored in sheets/docs
- Employees ask manually in chat/email
- HR/admin searches or replies repeatedly
- Answers may be delayed or inconsistent

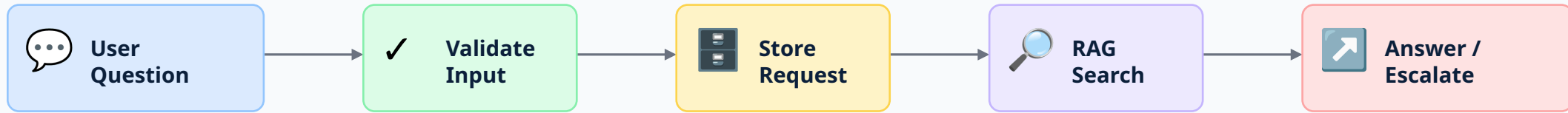
Pain Points

- Repetitive manual work
- Slow response time
- Risk of outdated or guessed answers
- No consistent fallback when information is missing

Goal: a reliable assistant that answers from approved knowledge only — with sources and escalation when needed.

Proposed Solution

A Company Knowledge AI Assistant built with n8n, Gemini, and Supabase vector search.



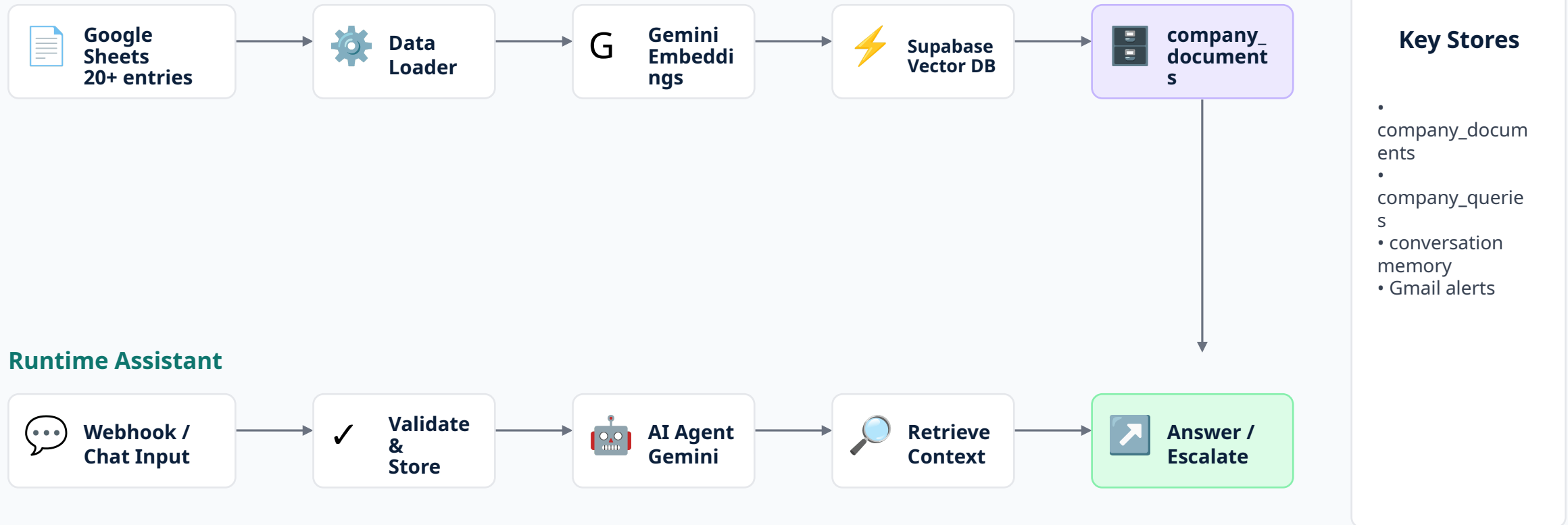
Structured statuses: answered · fallback · escalate

The assistant can answer known policies, safely say when information is unavailable, and notify a human for review cases.

System Architecture

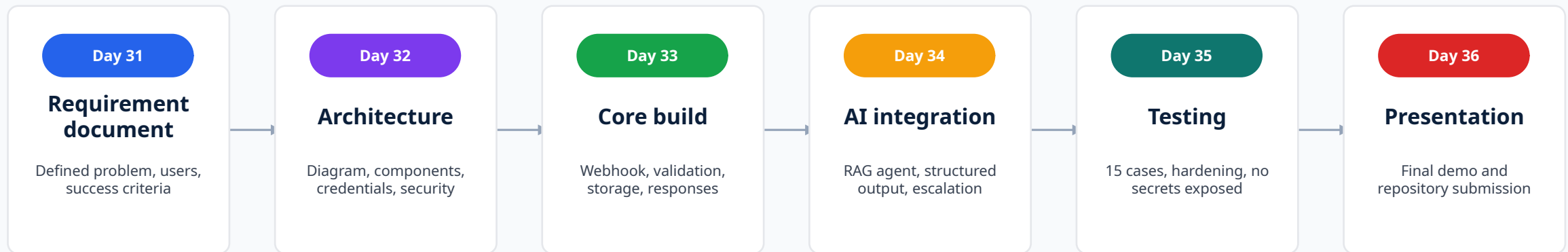
Two connected layers: knowledge ingestion and runtime question answering.

Knowledge Ingestion



Implementation Progression

The capstone was built incrementally across project days to reduce risk.



This approach kept the working Day 24 RAG assistant as the knowledge foundation, then layered deterministic validation and escalation around it.

AI Integration & RAG Logic

AI is only used where retrieval, reasoning, and natural-language generation are needed.

AI Agent

- Search knowledge base first
- Answer only from retrieved context
- Use memory for follow-up questions
- Return structured status

RAG Guardrails

- No general-knowledge policy guesses
- Unknown → fallback response
- Source line required when available
- Human review for sensitive cases

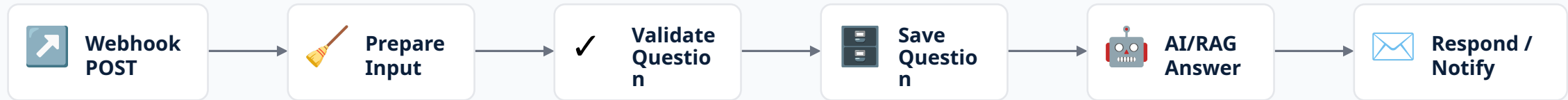
Structured Output

```
{  
  "answer": "...",  
  "source": "...",  
  "status": "answered | fallback |  
  escalate"  
}
```

The workflow routes based on status instead of trusting free-form text.

Core Workflow & Business Rules

Deterministic automation protects the AI layer from bad inputs and records requests.



Validation Rules

Required field · text conversion · length $\leq$ 500 · invalid input rejected before AI

Storage Rules

company_queries stores question, status, ID, and timestamp for tracking and auditing.

Routing Rules

answered/fallback returns to user; escalate triggers Gmail notification and webhook response.

Reliability, Error Handling & Security

The final workflow was hardened for invalid inputs, service failures, and credential safety.

Failure Handling

- Invalid/missing input rejected safely
- Fallback for unknown knowledge
- Gemini/API failure tested
- Supabase/database failure tested
- Human review notification for escalation

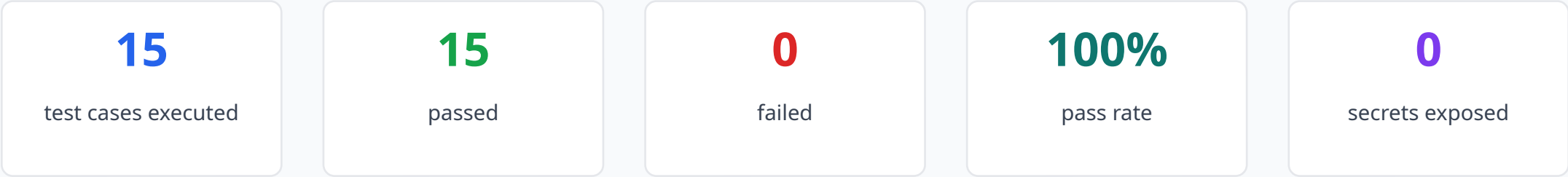
Security Controls

- Credentials stored in n8n credential vault
- No API keys hard-coded in workflows
- Workflow JSON inspected for secrets
- Source-aware answers avoid policy hallucination
- Optional authentication planned for production

Core reliability principle: fail safely, never invent answers, never expose secrets.

Testing & Optimization Results

Day 35 full test pass: 15 scenarios across normal, invalid, failure, security, and memory cases.



Coverage Summary



PASS

Final hardened workflow JSON exported after all tests passed.

Live Demo Plan

A focused demo that proves retrieval, fallback, escalation, validation, and memory.

- 1 Known policy** Ask: "How many casual leaves are allowed?" → source-aware answer
- 2 Fallback** Ask: "What is the maternity leave policy?" → information unavailable
- 3 Escalation** Ask formal complaint request → Gmail review alert fires
- 4 Validation** Send empty / oversized question → rejected before AI
- 5 Memory** Ask follow-up: "What about annual leaves?" → context-aware response

If the Gemini rate limit is hit during recording, show the saved screenshots and test report as execution evidence.

Business Benefits & Value

A practical assistant that reduces repetitive internal support work while improving answer quality.



Faster answers

Employees get immediate responses for common policy questions.



Reduced manual effort

HR/admin teams spend less time repeating the same answers.



Source-aware trust

Responses include policy/source references where available.



Safe fallback

Unknown questions are not invented or guessed.



Human escalation

Sensitive cases are routed to people instead of over-automated.



Auditable data

Questions are stored for analysis and future knowledge gaps.

Limitations & Future Improvements

The current version is functional, but production use needs stronger controls and broader knowledge coverage.

Current Limitations

- Knowledge base is a 20-entry sample dataset
- Policy updates require re-ingestion
- No production user authentication yet
- External Gemini/Supabase availability affects response time

Future Improvements

- Add real approved company documents
- Role-based access and user authentication
- Automated re-ingestion on document updates
- Telegram/Teams interface and analytics dashboard

Final Submission Package

Everything is organized in GitHub with day-wise documentation and final capstone deliverables.

Workflows

Final hardened workflow JSON + previous WIP exports

Documentation

Requirements, architecture, components, README, test cases

Knowledge source

20+ entry company knowledge base used for ingestion

Evidence

Screenshots for Q&A, fallback, escalation, tests, demo

Final takeaway

The capstone delivers a complete RAG/agent assistant with input validation, request storage, grounded answers, fallback, escalation, security checks, and full test coverage.

Thank you